

Microbiology: Detection of Microbes

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Session Objectives

1. Explain the basic and general aspects of diagnostic/clinical microbiology of the microbial world & their role in causing diseases.
2. Describe the various test procedures that are often used in a clinical microbiology lab for the purpose of identifying the etiologic agent(s) and for aiding the clinician in making the correct diagnosis.
3. Compare and contrast the different types of diagnostic microbiological assays
4. Determine diagnosis of infectious diseases based on laboratory findings

Lecture Outline

1. Basic Laboratory Tests

2. Identification by microscopy

1. Gram Stain
2. Auramine/Rhodamine Stain
3. Giemsa stain
4. Capsule, Flagellar & Spore stains

3. Identification by culture

1. Blood, Chocolate, MacConkey/EMB, Lowenstein/Jensen agar, Sabouraud, Bordet-Gengou, etc.

4. Identification by biochemical testing/Antigen Detection/Serology/Molecular techniques

Laboratory Tests

- 1) Routine blood tests (e.g. CBCs)
- 2) Staining / Microscopy
- 3) Culture
- 4) Biochemical Testing
- 5) Antigen Testing (e.g. ELISA, Latex Agglutination)
- 6) Serology (e.g. direct and indirect IFA, Western blots)
- 7) Molecular techniques (NAA/PCR; gene probes)

** It is noteworthy that culturing a specimen onto agar plates is still the most common way of identifying the causative pathogen.*

*** Serology, Antigen detection and molecular techniques are used often when cultures cannot be done. (why can't cultures be done?)*

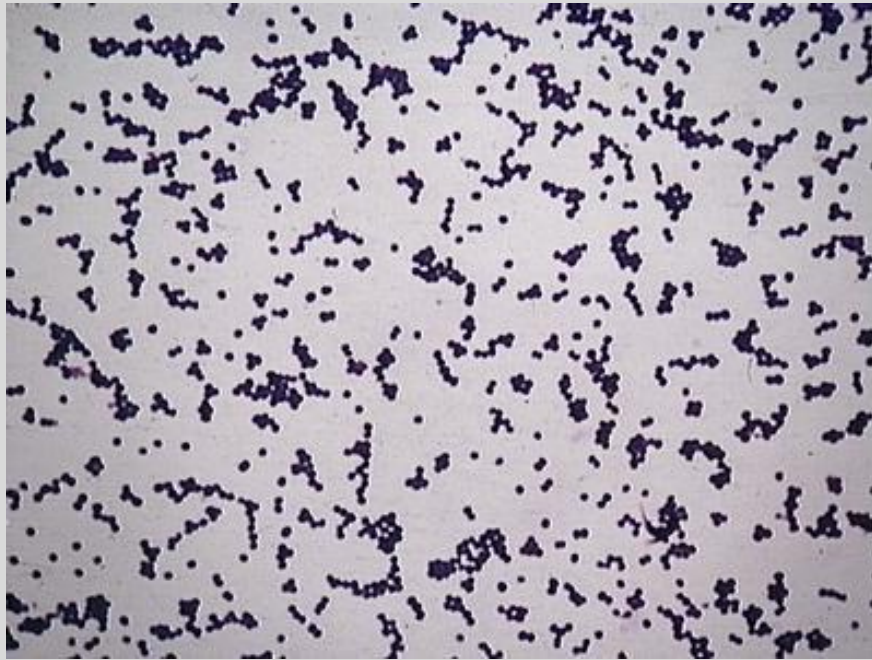
Microscopy

1. Gram stain: Gram-positive (purple-colored) vs. Gram-negative (red or pink). [most common bacteriologic stain]
2. Auramine/Rhodamine stain: helps identify *Mycobacteria* species and *Cryptosporidium* (a protozoan) – these will appear as **red-** or **maroon**-colored organisms (potassium permanganate counterstain). *Replaced the previously used acid-fast stain or Ziehl-Neelsen stain*
3. Wright-Giemsa stain: most commonly used for staining peripheral blood smears to identify protozoan parasites, esp. *Plasmodia* (malaria), *Babesia*, *Trichomonas*, *Toxoplasma* & some bacteria, such as *Chlamydia* and *Rickettsia*.
4. India Ink Stain: very useful for identifying *Cryptococcus neoformans*, mostly from a CSF sample

Microscopy Cont.

1. Capsule stain: identifies bacteria (and one fungi) that have capsules – e.g. *Streptococci*, *Cryptococci*, etc.
2. Flagellar stain: identifies motility
3. Spore stain: identifies spore-forming bacteria – e.g. *Bacillus & Clostridia*

Gram stain



← *Staph. aureus*: Gram + (appear as **clumps**)

E. coli: Gram-neg. rod →

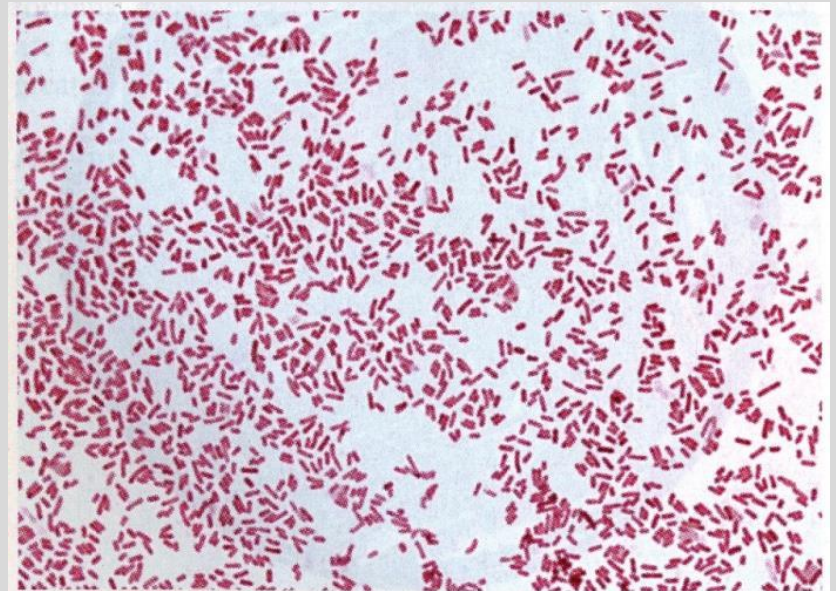
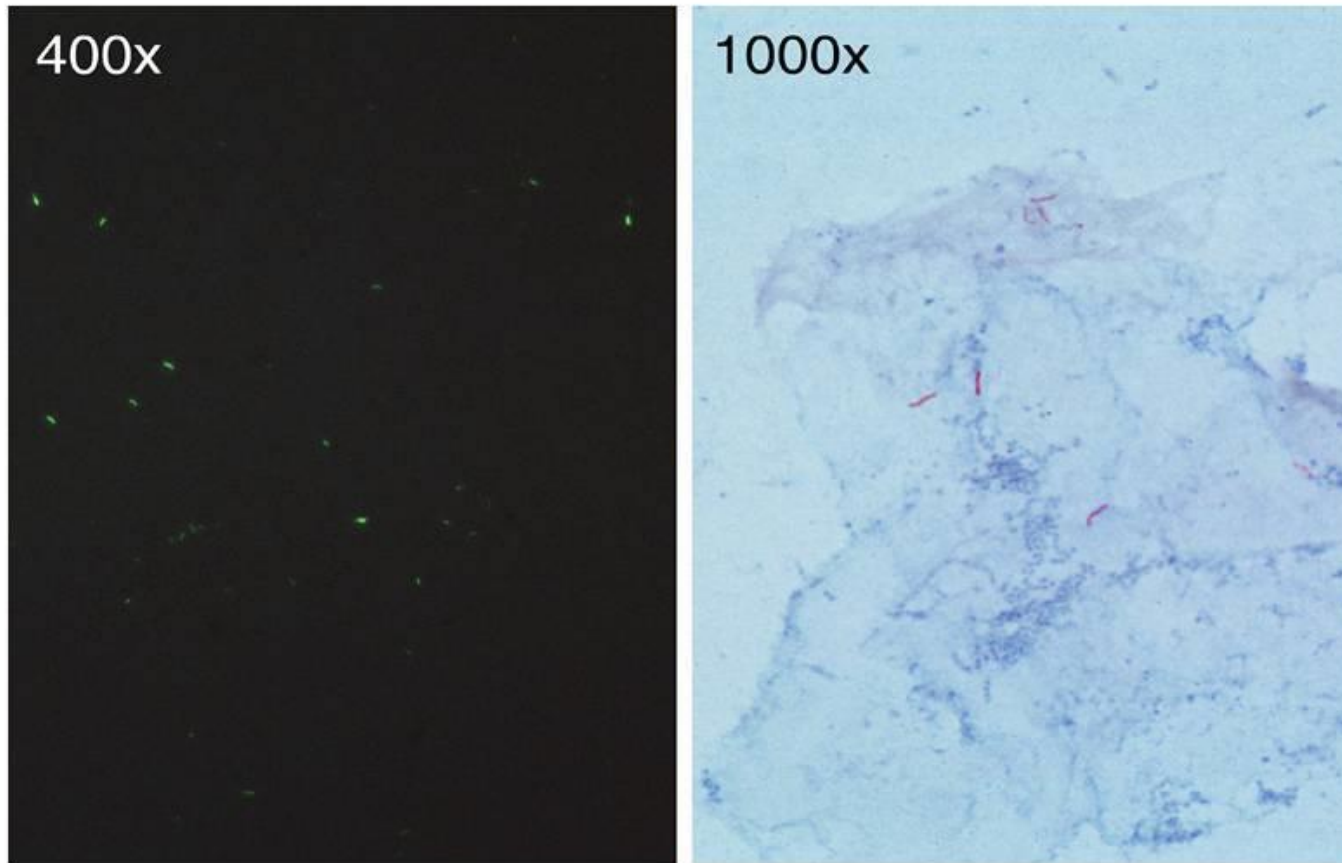


Figure 13-26. Gram Stain of *Escherichia coli* (X1000). Cells are usually arranged singly or in pairs. All members of *Enterobacteriaceae* are straight Gram-negative rods.

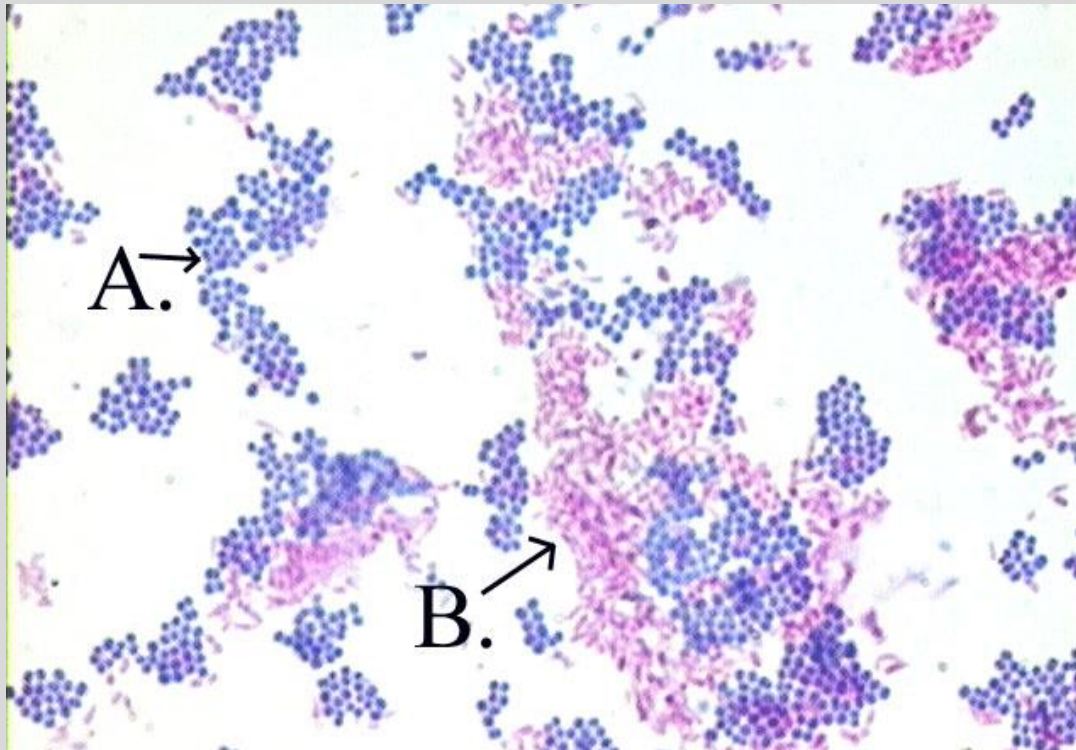
Auramine/Rhodamine stain



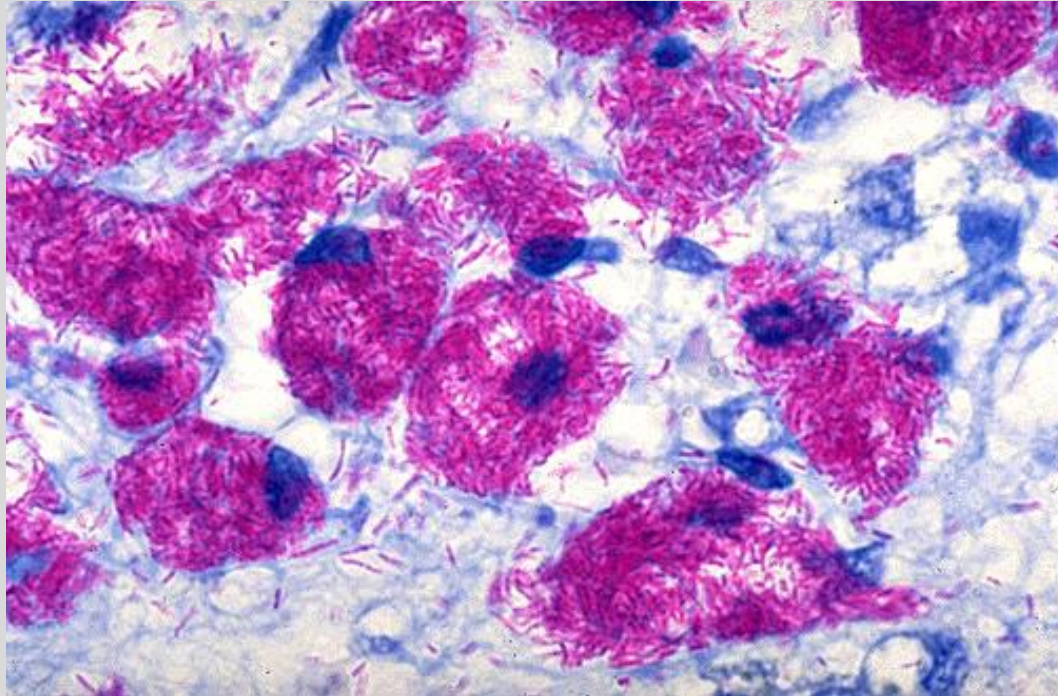
Source: Division of Laboratory Systems,
Public Health Practice Program Office, Center for Disease Control and Prevention (CDC)

Acid-fast stain

“A” points to **negative** cells (stain blue);
“B” points to **positive** bacteria (stain maroon)



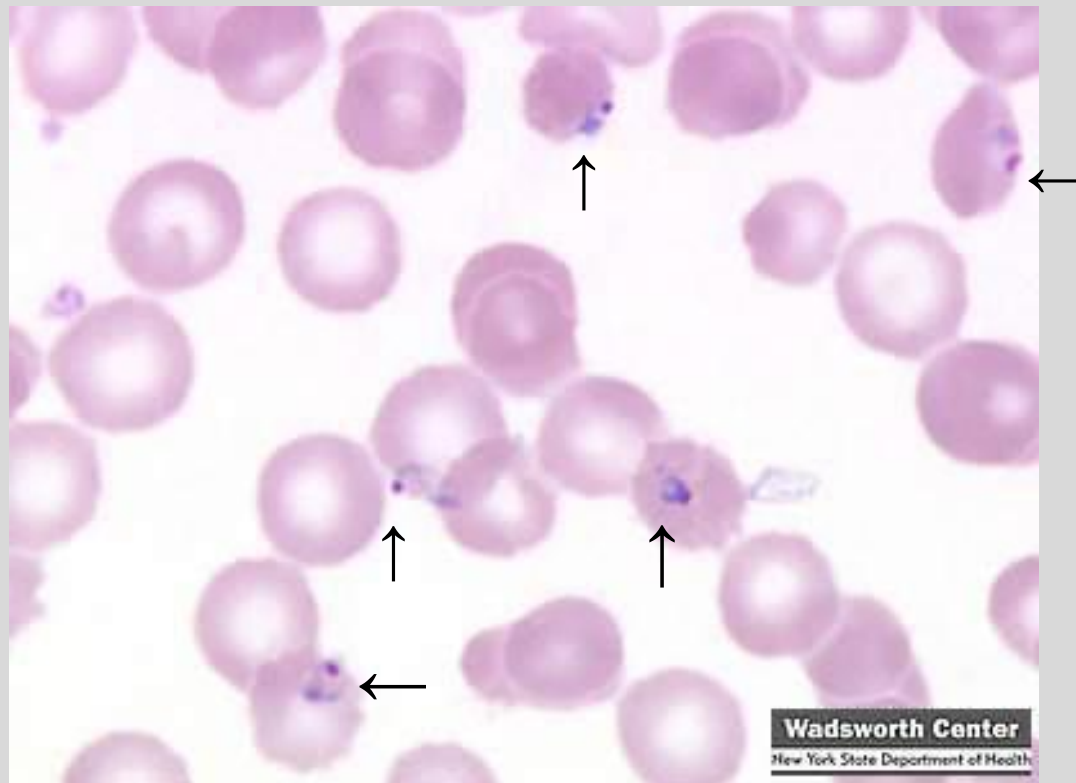
Acid-fast stain continued



← numerous acid-fast
Mycobacteria can be seen
inside (cytoplasm)
macrophages

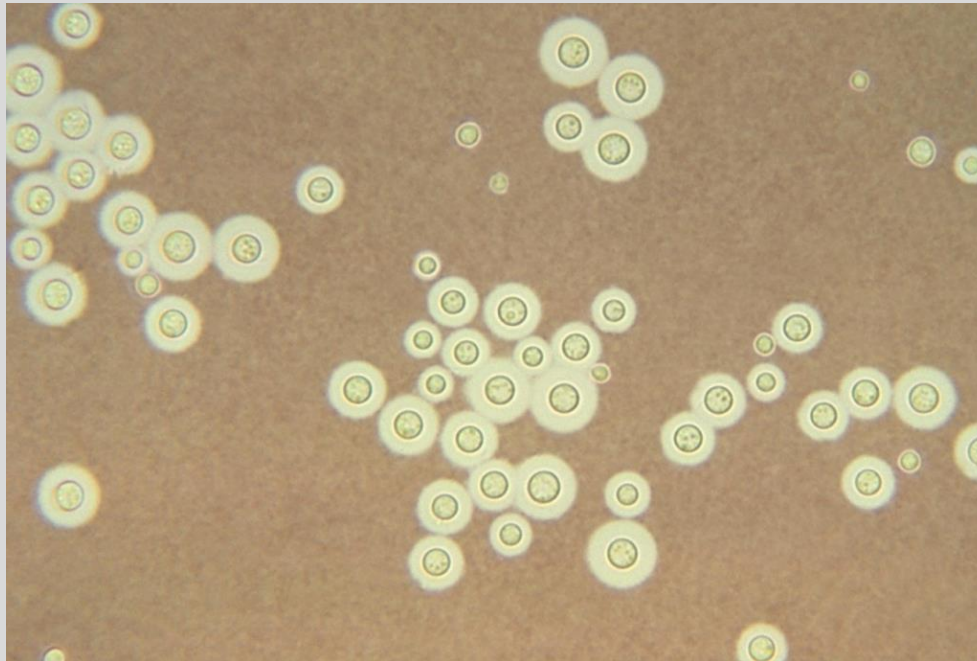
Wright-Giemsa stain

Giemsa stain of a peripheral blood smear showing 5 infected RBCs (with malaria parasites).

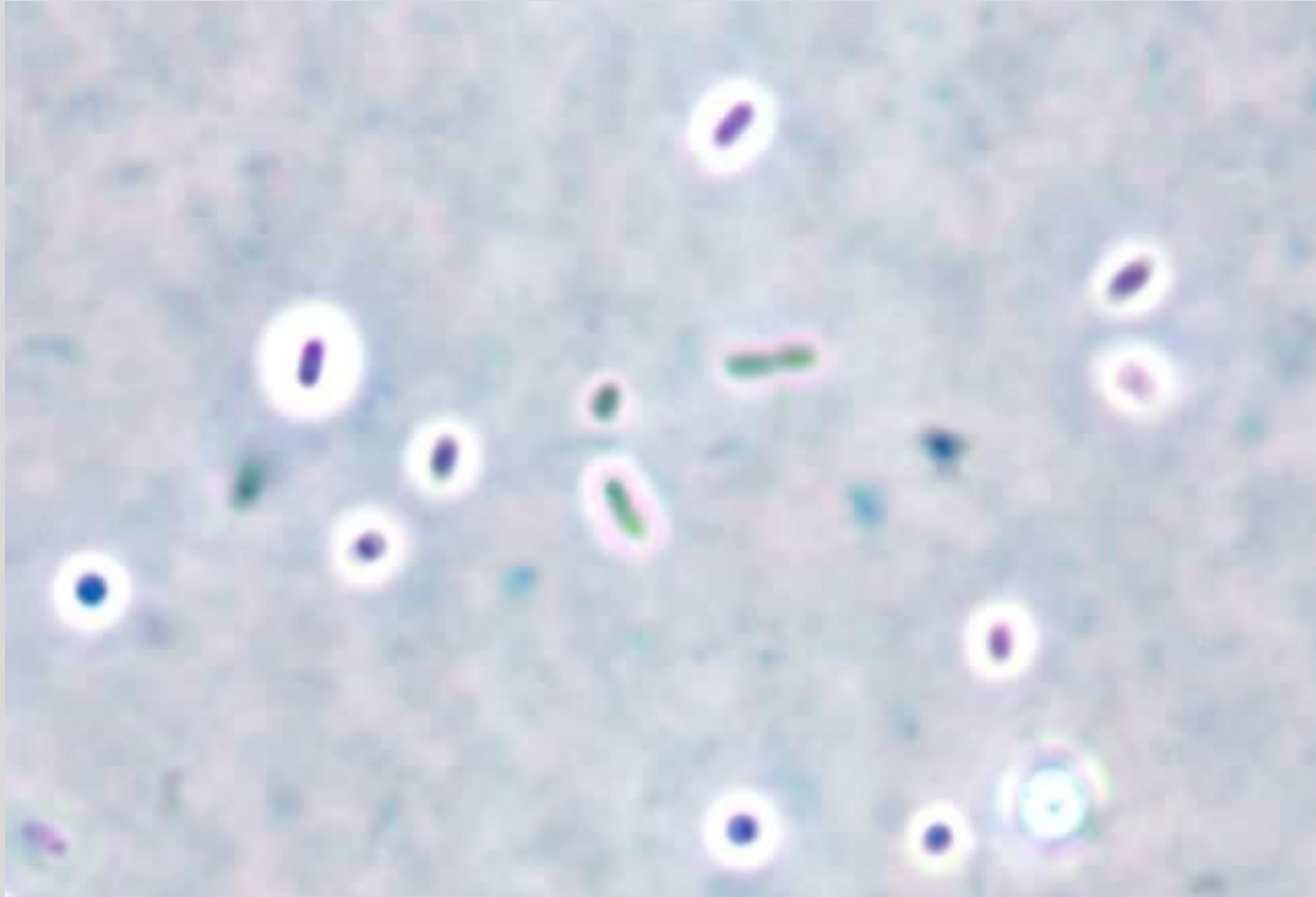


India Ink Stain

- **India ink** capsule stain of *Cryptococcus neoformans*



Capsule Stain



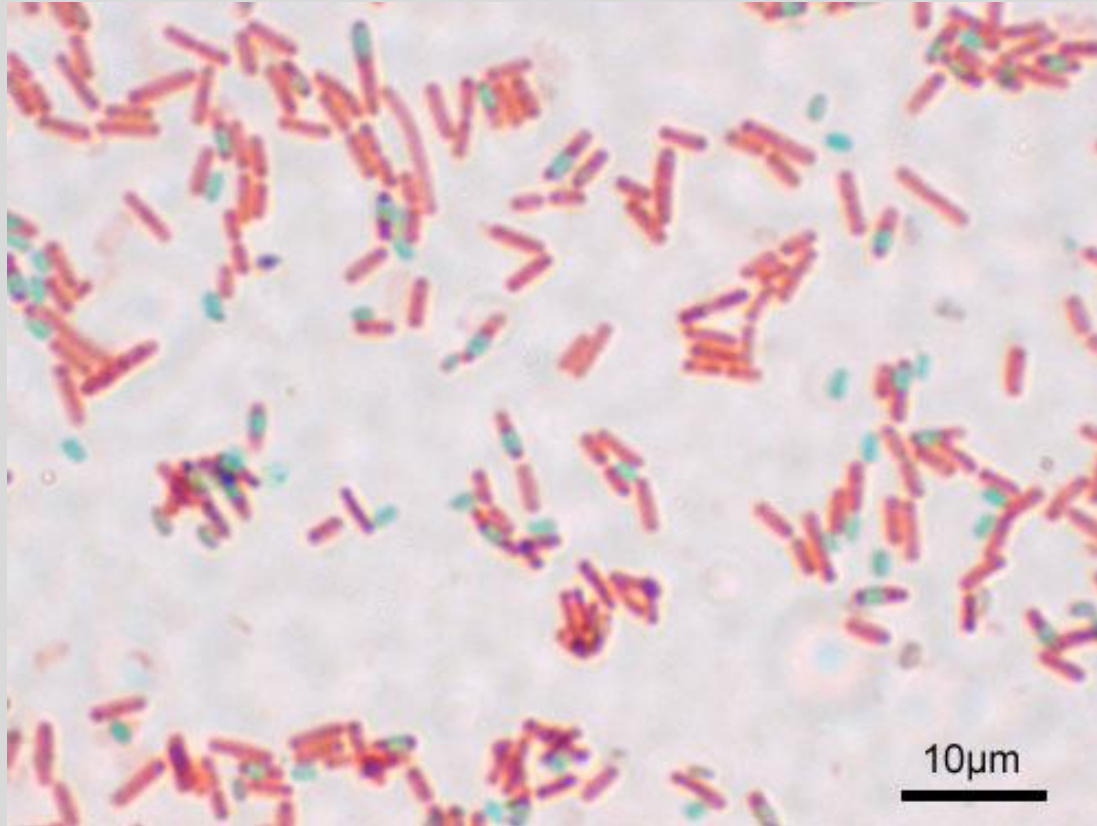
<http://microbeonline.com/wp-content/uploads/2016/10/Capsule-Stain.jpg>

Flagellar Stain



https://www.researchgate.net/figure/Flagellar-stain-of-a-Salmonella-Typhi-Like-E-coli-Salmonella-are-motile-by-means-of_fig26_290911856

Endospore Stain



https://bio.libretexts.org/Bookshelves/Microbiology/Microbiology_Laboratory_Manual_%28Hartline%29/01%3A_Labs/1.12%3A_Endospore_Stain

Identification based on culture

- **Source of culture material:**
 - Most common
 1. Blood
 2. Urine
 3. Stool
 4. CSF
 - Other
 1. skin (lesion or rash)
 2. sputum
 3. throat
 4. abscess
 5. synovial fluid
 6. biopsied material (from a specific tissue site or organ)

Culture Media Types

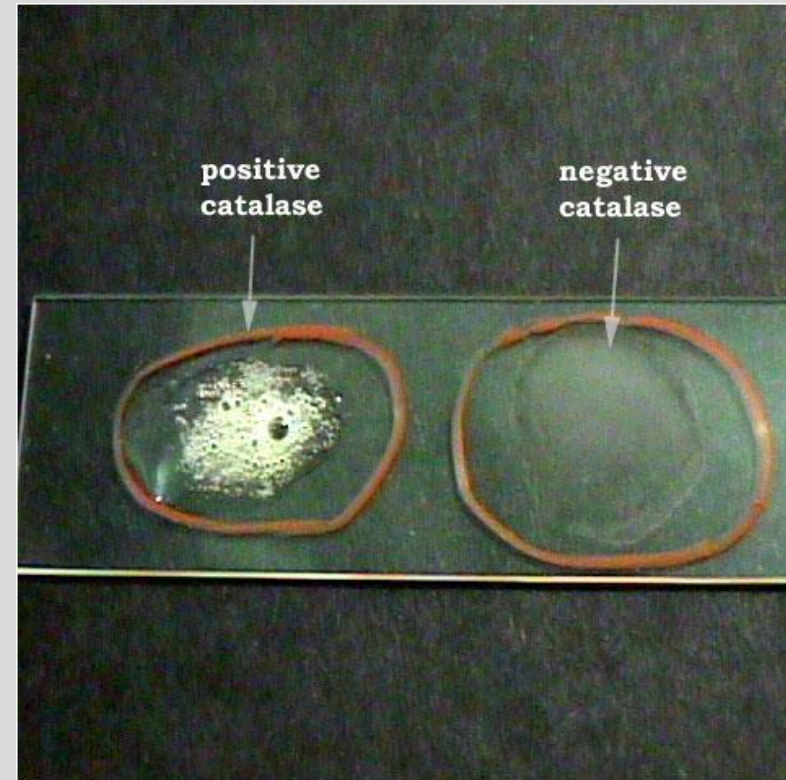
- Broth/Slant/Plates
- Agar Plates:
 1. Non-selective:
 - a) Blood agar – for growing out most non-fastidious bacteria
 - b) Chocolate agar – helps promote the growth of fastidious bacteria, esp. *Haemophilus influenzae* and *Neisseria gonorrhoeae* (gonococcus)
 - c) Sabouraud Dextrose agar – promotes the growth of most fungi
 2. Selective/differential:
 - a) MacConkey / EMB agar – Gram -ve **enteric** rods (family *Enterobacteriaceae*); lactose-fermenters vs. non-fermenters
 - b) Lowenstein/Jensen agar – promotes the growth of *Mycobacteria*, esp. *M. tuberculosis*
 - c) Buffered charcoal yeast extract – for *Legionella pneumophila*
 - d) Bordet Gengou agar – *Bordetella pertussis*
- Tissue cell lines for viruses

Culture conditions

- O₂ requirements –
 - a) Aerobic
 - b) Anaerobic
 - c) Microaerophilic ($\leq 5\%$)
- Temperature –
 - a) at or near room temp.: $\sim 25^{\circ}\text{C}$ (only best for some bacteria/ most fungi);
 - b) at or near body temp.: $\sim 35\text{-}37^{\circ}\text{C}$ (ideal for most microorganisms);
 - c) at above body temp.: $\sim 40\text{-}42^{\circ}\text{C}$ (ideal for very few)

Biochemical Assays

- **Variety** (will discuss key ones)
- **Catalase test:**
 - a) for distinguishing *Staphylococci* spp. (catalase+) from the *Streptococci/Enterococci* (catalase-neg)
 - b) These 2 groups can be indistinguishable morphologically because they appear as Gram-positive cocci of about the same size and shape albeit different pattern



Biochemical Assays

Coagulase test

- Coagulase test – *Staphylococcus aureus* (a common pathogen) is **coagulase +ve**, whereas most other *Staph. spp.*, such as *S. epidermidis* and *S. saprophyticus* are **coagulase -ve**



← Clumping of plasma (+)

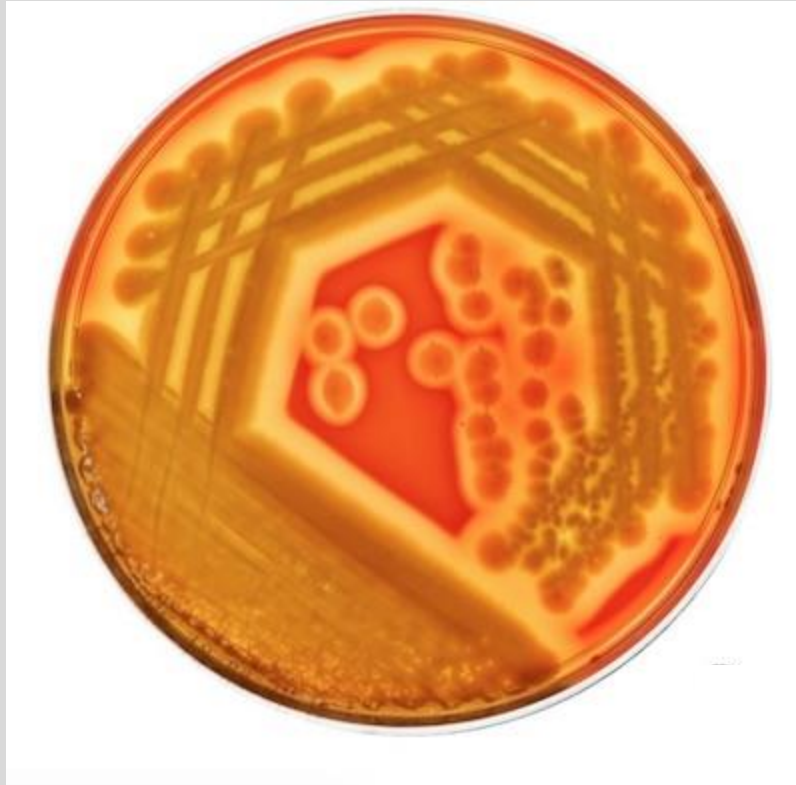
← No clumping (negative)

Biochemical Assays

Hemolysis

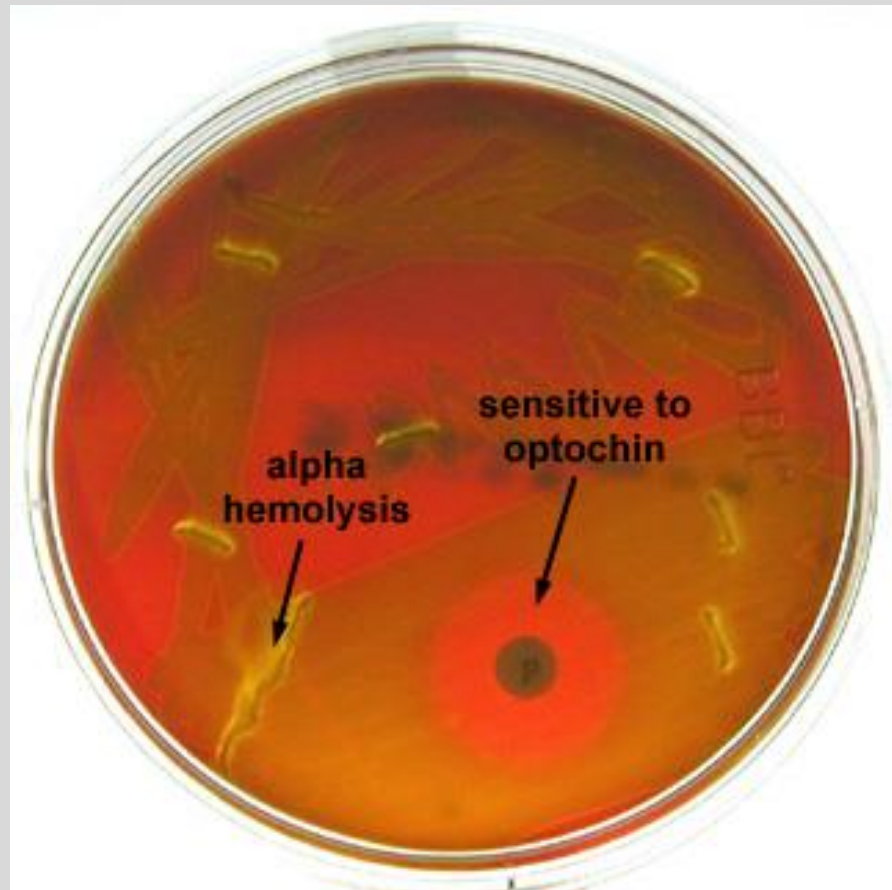
- **Beta-hemolysis** – some bacteria are capable of completely hemolyzing blood
 - a) Colonies have a clear zone of hemolysis on blood agar
 - b) E.g. all Group A and Group B *Streptococci*, most strains of *S. aureus*, some strains of *Clostridia*, and about half of *E. coli*.
 - c) Hemolysis is caused by an enzyme hemolysin
- **Alpha-hemolysis** – some bacteria are capable of partially hemolyzing blood.
 - a) Colonies have a greenish or greenish-brown zone of hemolysis on blood agar
 - b) E.g. Various *Streptococci*, including *S. pneumoniae*, and are part of a much larger group of alpha-hemolytic *Streptococci* known as the viridans *Streptococci*
- **Gamma-hemolysis** – no hemolytic reaction on blood agar.
 - Colonies do not have a zone of hemolysis
 - E.g. *Enterococcus faecalis* (UTIs), *S. saprophyticus*, *S. epidermidis*

Example of Beta hemolysis



<https://microbialmenagerie.com/blood-agar-hemolysis/>

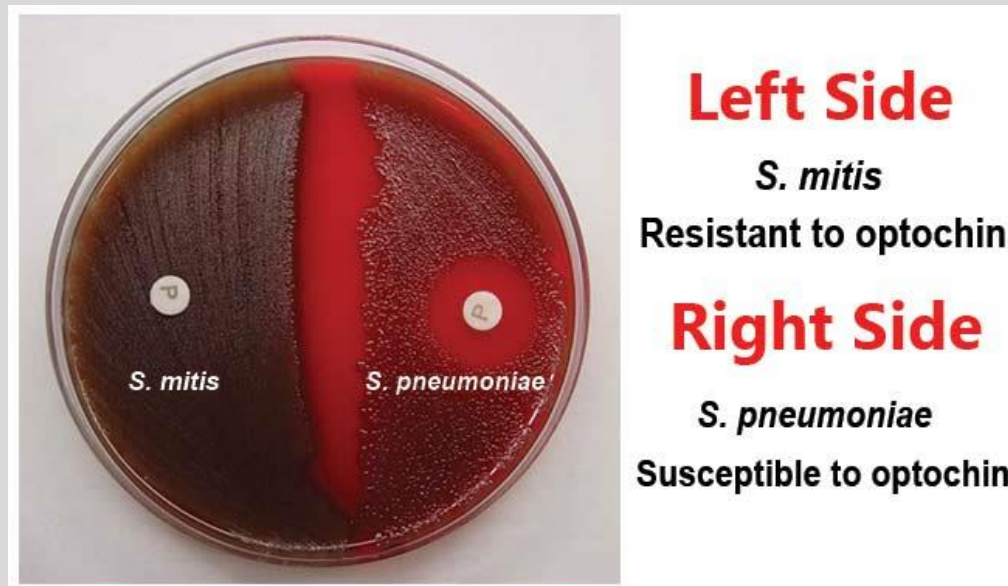
Example of alpha hemolysis caused by *Streptococci*



Biochemical Assays

Disc sensitivity testing

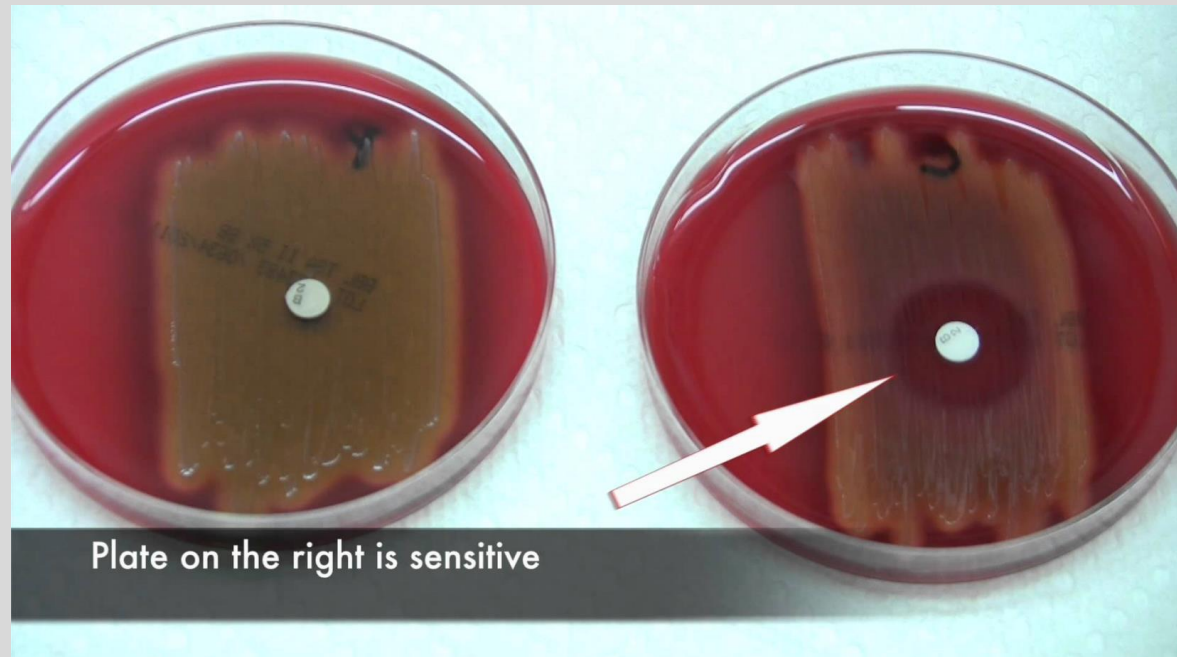
- P-disc sensitivity – *S. pneumoniae* (pneumococcus) does not grow in the presence of the “P” – disc (or “O” disc), which has a selective inhibitor, **optochin**, incorporated into it.
- All other **alpha**-hemolytic *Streptococci* are resistant



Biochemical Assays

Disc sensitivity testing

- A-disc sensitivity – Group A *Streptococci* (aka *S. pyogenes*) inhibited by “A”-disc, which has the antibiotic, **bacitracin**.
- All other **beta**-hemolytic *Streptococci* are resistant to the A-disc.



Biochemical Assays

Sugar Fermentation

- **Lactose** fermentation – agars like MacConkey, have lactose incorporated into them. If the bacteria utilize lactose, the agar will change color (usually to red or purple)
- Most strains of *E. coli* and *Klebsiella* are **lactose-fermenters**, whereas most other Gram –ve rods, such as *Salmonella* & *Shigella* are not. *Salmonella* & *Shigella* will form pale or colorless colonies instead.

Lactose fermentation: MacConkey agar



← uninoculated plate

Lactose-negative

Lactose-positive



Biochemical Assays

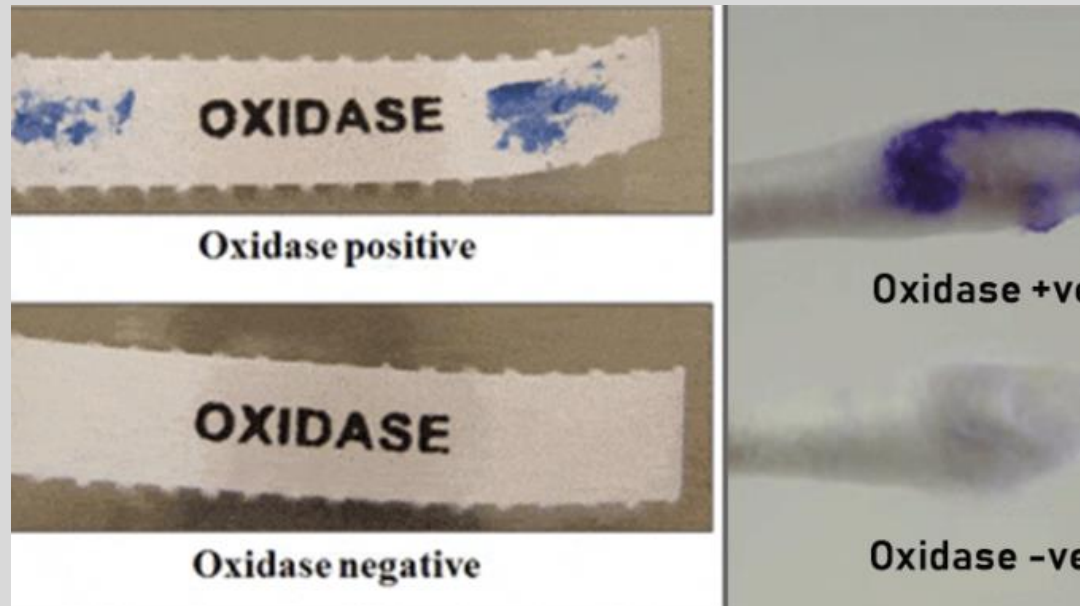
Sugar Fermentation

- **Sorbitol** fermentation – another key sugar, sorbitol, can be incorporated into MacConkey agar for the purpose of identifying perhaps the most serious *E. coli* of them all, *E. coli* **0157**, also known as enterohemorrhagic *E. coli* (**EHEC**) or Shiga-toxin producing *E. coli* (**STEC**).
- It causes serious, life-threatening kidney failure, mostly in young children. EHEC will **not** ferment sorbitol, whereas most other non-EHEC strains of *E. coli* will ferment sorbitol.

Biochemical Assays

Oxidase Reaction

- **Oxidase** activity – some bacteria possess the enzyme oxidase (ETS).
 - E.g.: *Pseudomonas*, *Campylobacter*, *Helicobacter*, *Vibrio*, *Neisseria*, & *Pasteurella*.



<https://www.onlinebiologynotes.com/oxidase-test-principle-procedure-result-interpretation-precautions/>

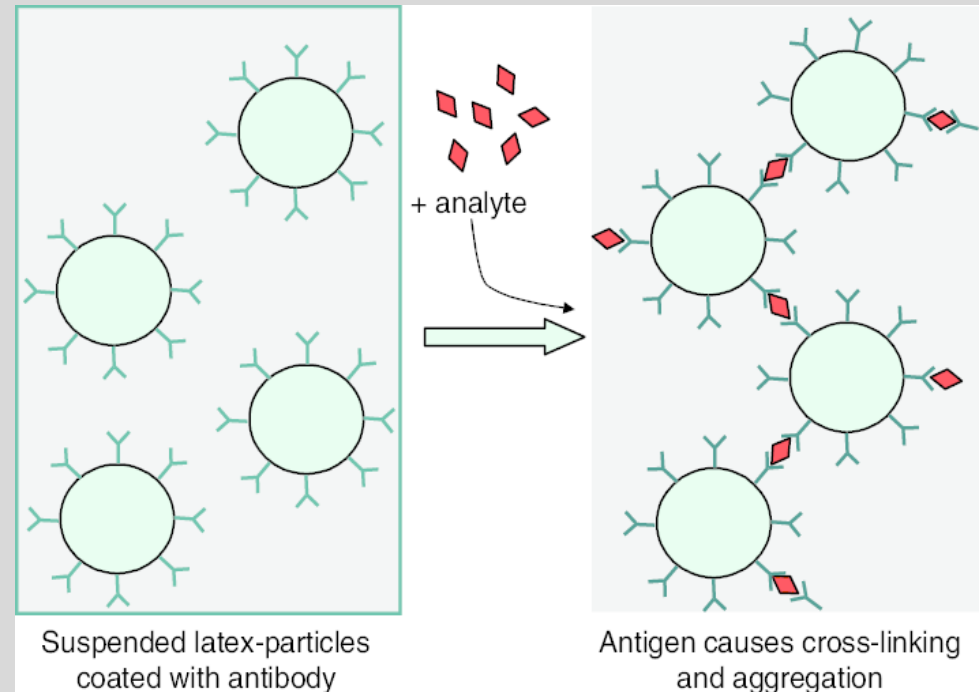
Antigen Detection

- Antigen detection: immunologic systems designed to detect microbial antigens
 - a) Rapid tests (mins)
 - b) Built in controls
- Examples:
 1. ELISA
 2. Latex Agglutination tests

Antigen Detection

Latex Agglutination

- Latex Agglutination tests: Ab fixed to latex beads; Ag binds to Ab → agglutination of beads; Streptococcal pharyngitis, Cryptococcal Ag

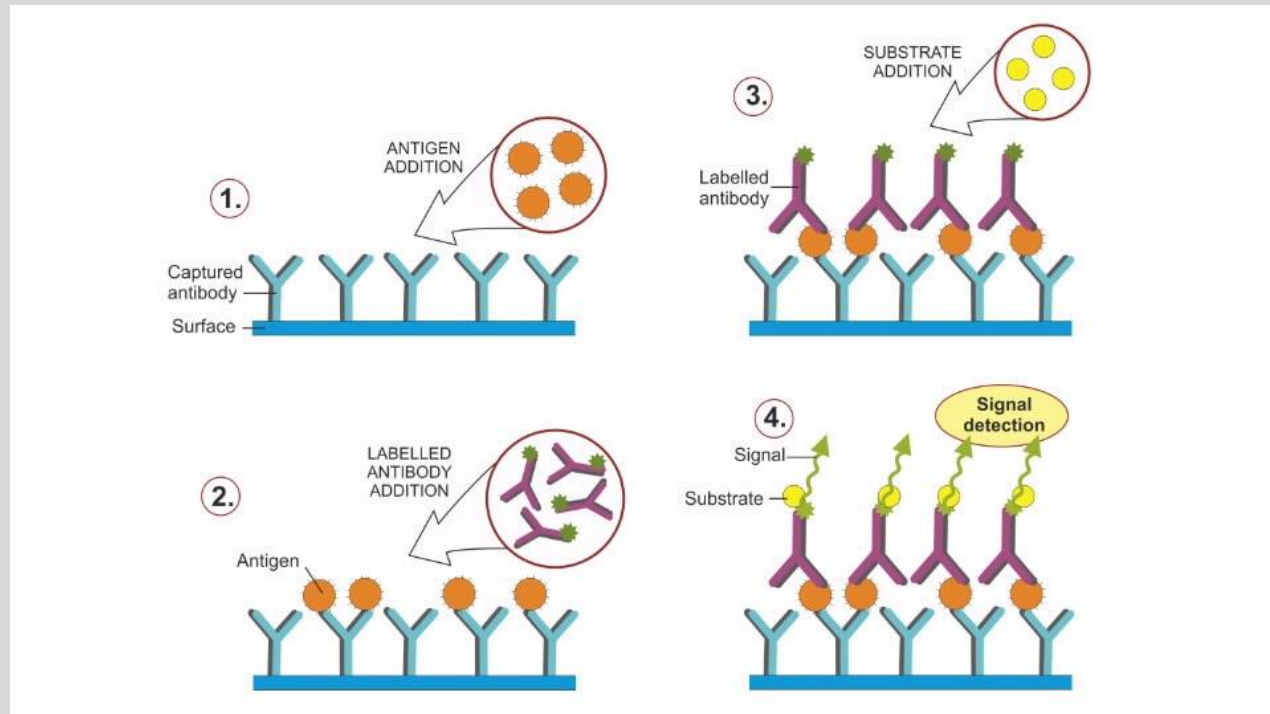


https://www.researchgate.net/figure/Latex-agglutination-assay-The-latex-agglutination-assay-is-a-classical-immunochemical_fig1_287275188

Antigen Detection

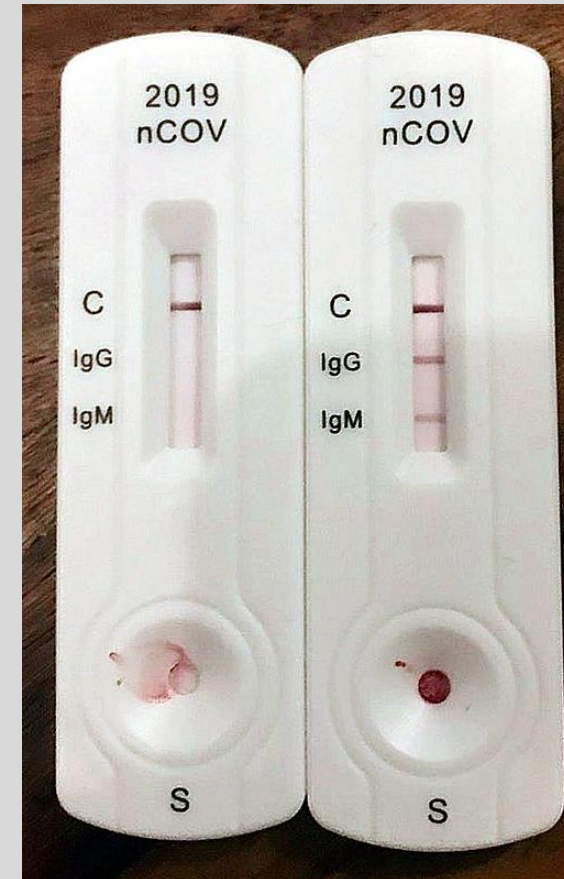
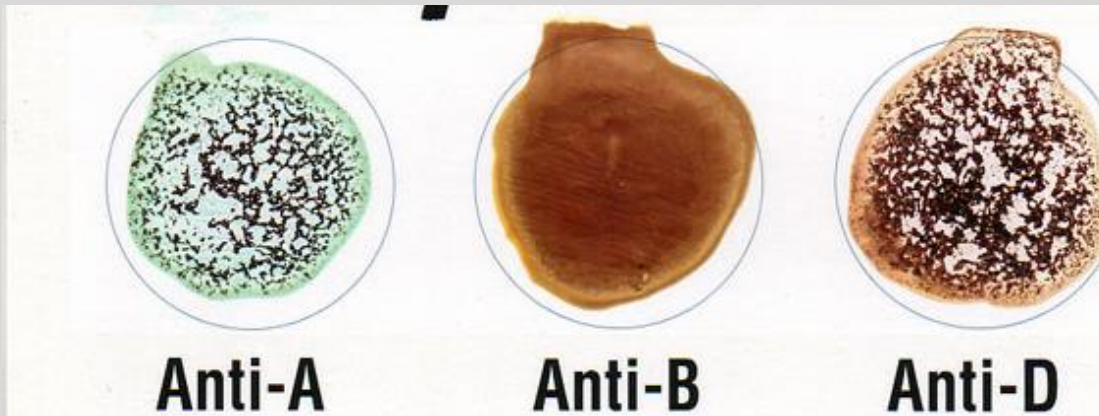
ELISA

- ELISA
 - Color change
 - Conc of product generated is proportional to the quantity of Ag
 - E.g. Autoantibodies such as anti-dsDNA, Antinuclear Abs, Hepatitis A, B, and C, HIV, PSA, LH, FSH, Human chorionic gonadotropin H



Serology

- Serology: The detection of **antibodies** and other substances in serum.
- Several assays such as agglutination, direct immunofluorescence, complement fixation, Western blot and ELISA
- Used for a lot of fungal infections such as Coccidioidomycosis, Histoplasmosis, Blastomycosis, Sporotrichosis,



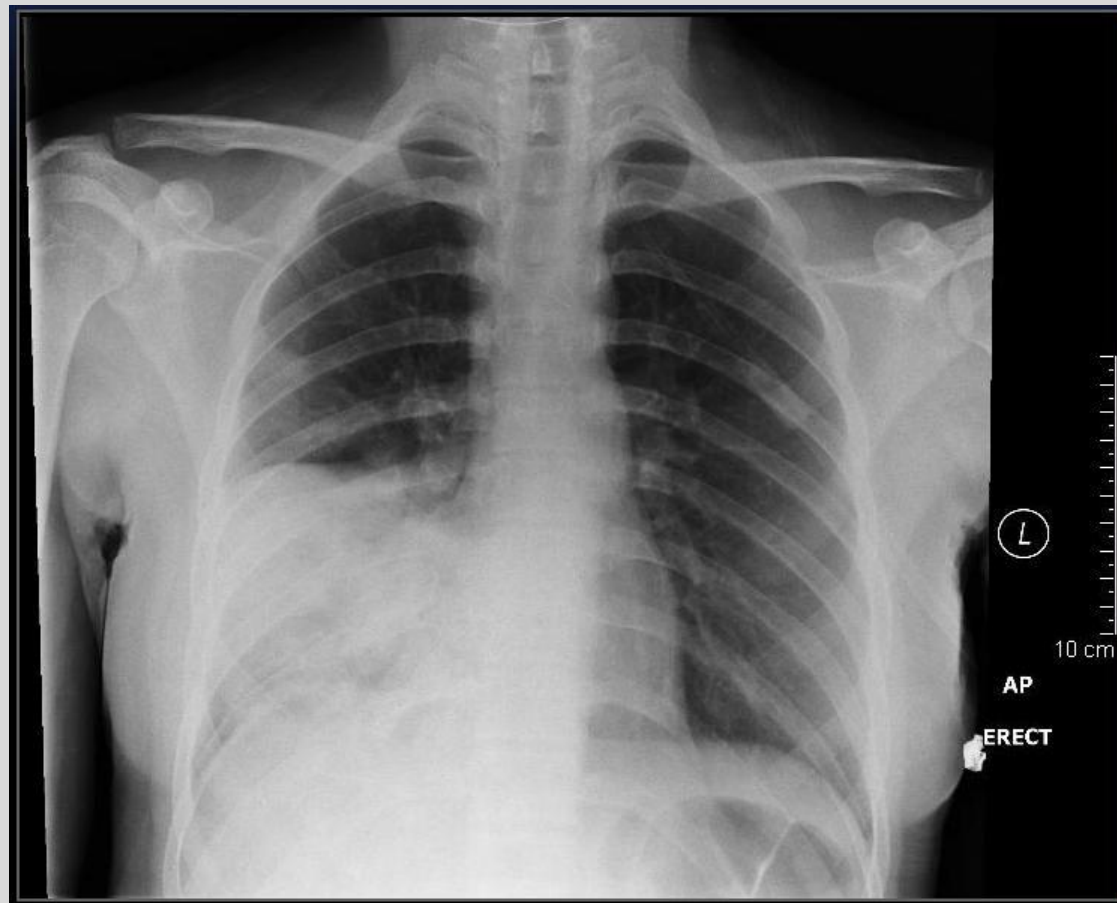
Molecular Techniques

- Detection of macromolecules (DNA/RNA/proteins)
- Rapid, high specificity & sensitivity
- Excellent alternative when microbe can't be cultured
- Types:
 - Hybridization Probes (FISH): detect several bacteria (*S. aureus*, *Enterococci*) and fungi (*Candida*)
 - Nucleic Acid Amplification (PCR and non-PCR based)
 - PCR: COVID-19, Influenza,
 - Non-PCR based: *Clostridioides difficile*

Practice Clinical Case

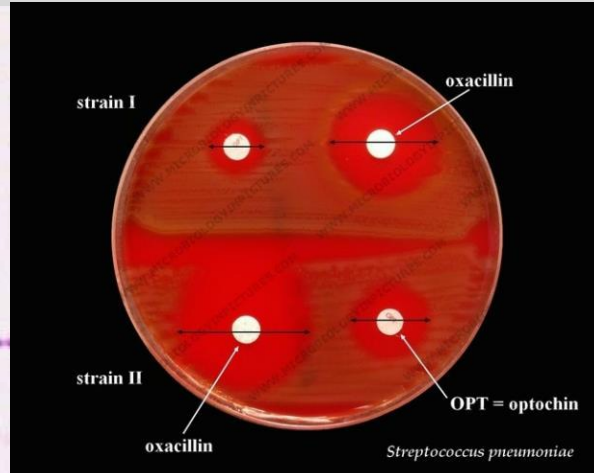
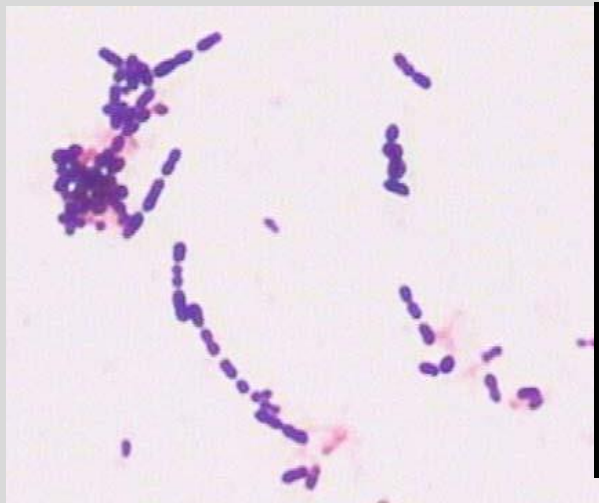
- A 70-year-old woman presented to the emergency department with a 2-day history of dyspnea and a cough productive of white phlegm. She denied hemoptysis, night sweats, fever, chills, abdominal pain, nausea, vomiting or chest pain. Medical history is pertinent for a diagnosis of multiple myeloma a year ago. She also had been treated with five cycles of immunosuppressive drugs including prednisone, with the last cycle completed 6 weeks ago. On physical examination, she had a temperature of 39.4 °C, HR of 120 beats/min, and respiratory rate of 20/min. Chest auscultation was significant for bilateral crackles with expiratory wheezes. Chest radiograph (next slide) showed mostly diffuse infiltrates with effusion of the right lower lobe.

Patient's chest X-ray



Clinical Case cont'd:

- White blood cell count was 1,700 cells per μl . She had a P_{O_2} of 33 mm of Hg which was corrected by receiving oxygen by nasal cannula. Three sets of blood cultures were obtained & she was started on i.v. cefotaxime & clindamycin for presumed bacterial pneumonia. Gram stain micrograph of the organism recovered from her blood is shown in Fig.1. Optochin disc sensitivity & susceptibility testing of the isolate are shown in Fig. 2. Catalase test is in Fig 3.



←Fig.2

← P disc



Fig.3

Questions related to the Case

1. What is the etiological agent of this pt's infection?
2. What risk factors does this woman possess for developing infection with this organism?
3. How does the catalase and optochin disc tests help distinguish this pathogen from other similar organisms?
4. What type of hemolysis reaction would this organism cause blood agar?

Questions related to the Case cont'd

5. What is(are) the major virulence factor(s) for this organism and their role in the pathogenesis of disease?
6. What other populations are at risk for infection with this organism? What can be done to try to prevent this?

Example of an exam-type question

- A 28-year-old female presents to her family physician with a temperature of 101.5F, and frequent and painful urination, of 3 days duration, which was followed by lower flank pain. Culturing of a urine sample on MacConkey agar is shown below. The likely etiologic agent is

- A. Chlamydia trachomatis*
- B. Staphylococcus aureus*
- C. Salmonella typhi*
- D. Escherichia coli*
- E. Pseudomonas aeruginosa*



Summary of Concepts

1. A vast array of laboratory tests exist to aid in the diagnosis of an infection caused by a pathogenic microbe.
2. **Culturing** a clinical sample onto various types of media & agars is still perhaps the most common technique for identifying a pathogen.
3. When culturing a patient sample cannot be performed or a potential pathogen is difficult to culture and isolate, then alternative assays such as **serologic techniques** (designed to detect antibodies, & sometimes antigens) are used.
4. **Molecular**-based techniques, such as **PCR** & gene probes, are good alternatives to the preceding ones, especially when they would not be appropriate or useful, and for confirming the suspected diagnosis in conjunction with other criteria.

Summary

1. Basic Laboratory Tests

2. Identification by microscopy

1. Gram Stain
2. Auramine/Rhodamine Stain
3. Giemsa stain
4. Capsule, Flagellar & Spore stains

3. Identification by culture

1. Blood, Chocolate, MacConkey/EMB, Lowenstein/Jensen agar, Sabouraud, Bordet-Gengou, etc.

4. Identification by biochemical testing/Antigen Detection/Serology/Molecular techniques

References

1. Murray, P.R., Rosenthal, K. S., and Pfaller, M.A. 2021. **Medical Microbiology**. Elsevier Inc.
2. Tao, LE, V. Bhushan and M. Sochat. 2024. **First Aid for the USMLE Step 1**; McGraw Hill Publication.
3. Engleberg, N. C., DiRita, V. and T.S. Dermody. 2013. **Schaechter's Mechanism of Microbial Disease**. Wolters Kluwer Health 5th ed. Lippincott Williams & Wilkins. ISBN: 978-0-7817-8744-4.

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>